Task1_1:

```
Loading dataset with 80/20 train-validation split...
Found 158 files belonging to 3 classes.
Using 127 files for training.
Found 158 files belonging to 3 classes.
Using 31 files for validation.
Classes found: ['Baetidae sp', 'Elmis sp', 'Sialis sp']
```

Finds classes and files

Task1_2:

```
# === STEP 2: Optimize dataset performance ===

AUTOTUNE = tf.data.AUTOTUNE

train_ds = train_ds.prefetch(buffer_size=AUTOTUNE)
val_ds = val_ds.prefetch(buffer_size=AUTOTUNE)
```

Changes performance of dataset

Task1_3:

Task1_4:

Model: "functional_1"

Layer (type)	Output Shape	Param #
input_layer_2 (InputLayer)	(None, 224, 224, 3)	0
sequential (Sequential)	(None, 224, 224, 3)	0
true_divide (TrueDivide)	(None, 224, 224, 3)	0
subtract (Subtract)	(None, 224, 224, 3)	0
mobilenetv2_1.00_224 (Functional)	(None, 7, 7, 1280)	2,257,984
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1280)	0
dropout (Dropout)	(None, 1280)	0
dense (Dense)	(None, 3)	3,843

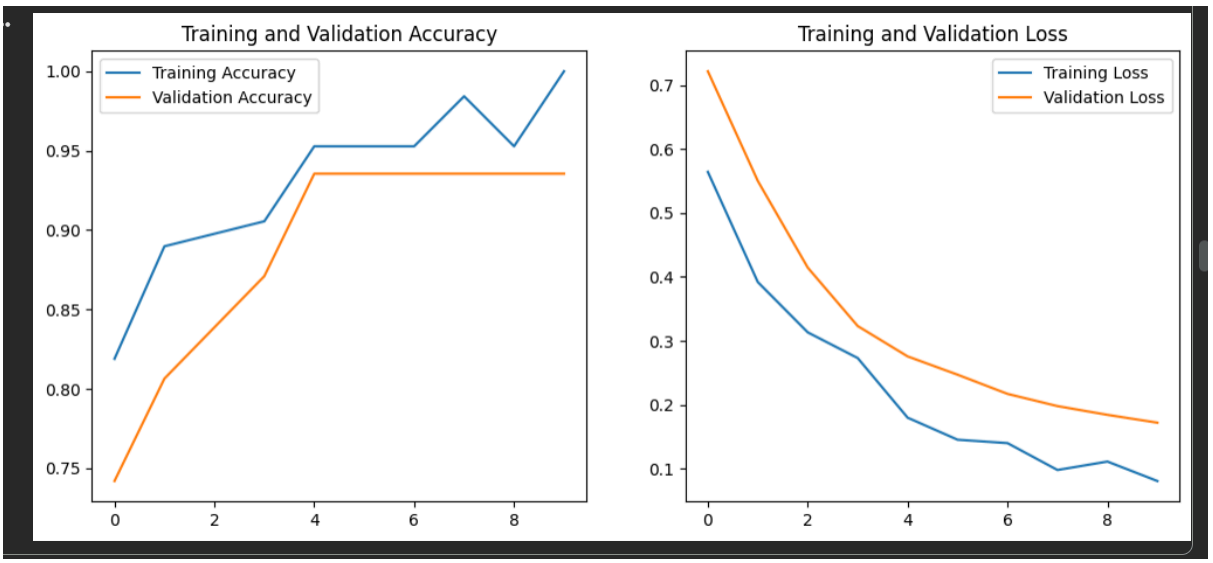
Total params: 2,261,827 (8.63 MB)
Trainable params: 3,843 (15.01 KB)
Non-trainable params: 2,257,984 (8.61 MB)

Task1_5:

```
Epoch 1/10
4/4 ━━━━━━━━━━━━━━━━━━━━ 11s 3s/step - accuracy: 0.8189 - loss: 0.5642 - val_accuracy: 0.7419 - val_loss: 0.7216
Epoch 2/10
4/4 ━━━━━━━━━━━━━━━━━━━━ 1s 382ms/step - accuracy: 0.8898 - loss: 0.3920 - val_accuracy: 0.8065 - val_loss: 0.5501
Epoch 3/10
4/4 ━━━━━━━━━━━━━━━━━━━━ 3s 388ms/step - accuracy: 0.8976 - loss: 0.3133 - val_accuracy: 0.8387 - val_loss: 0.4146
Epoch 4/10
4/4 ━━━━━━━━━━━━━━━━━━━━ 4s 666ms/step - accuracy: 0.9055 - loss: 0.2732 - val_accuracy: 0.8710 - val_loss: 0.3233
Epoch 5/10
4/4 ━━━━━━━━━━━━━━━━━━━━ 4s 399ms/step - accuracy: 0.9528 - loss: 0.1799 - val_accuracy: 0.9355 - val_loss: 0.2757
Epoch 6/10
4/4 ━━━━━━━━━━━━━━━━━━━━ 2s 401ms/step - accuracy: 0.9528 - loss: 0.1455 - val_accuracy: 0.9355 - val_loss: 0.2471
Epoch 7/10
4/4 ━━━━━━━━━━━━━━━━━━━━ 2s 407ms/step - accuracy: 0.9528 - loss: 0.1401 - val_accuracy: 0.9355 - val_loss: 0.2171
Epoch 8/10
4/4 ━━━━━━━━━━━━━━━━━━━━ 2s 399ms/step - accuracy: 0.9843 - loss: 0.0981 - val_accuracy: 0.9355 - val_loss: 0.1980
Epoch 9/10
4/4 ━━━━━━━━━━━━━━━━━━━━ 2s 406ms/step - accuracy: 0.9528 - loss: 0.1113 - val_accuracy: 0.9355 - val_loss: 0.1844
Epoch 10/10
4/4 ━━━━━━━━━━━━━━━━━━━━ 4s 652ms/step - accuracy: 1.0000 - loss: 0.0809 - val_accuracy: 0.9355 - val_loss: 0.1721
```

Displays accuracy, loss value accuracy and value loss

Task1_6:



Graphs training and Validation Accuracy and loss

Task1_7:

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 1s 540ms/step - accuracy: 0.9355 - loss: 0.1721
Validation accuracy: 93.55%
```

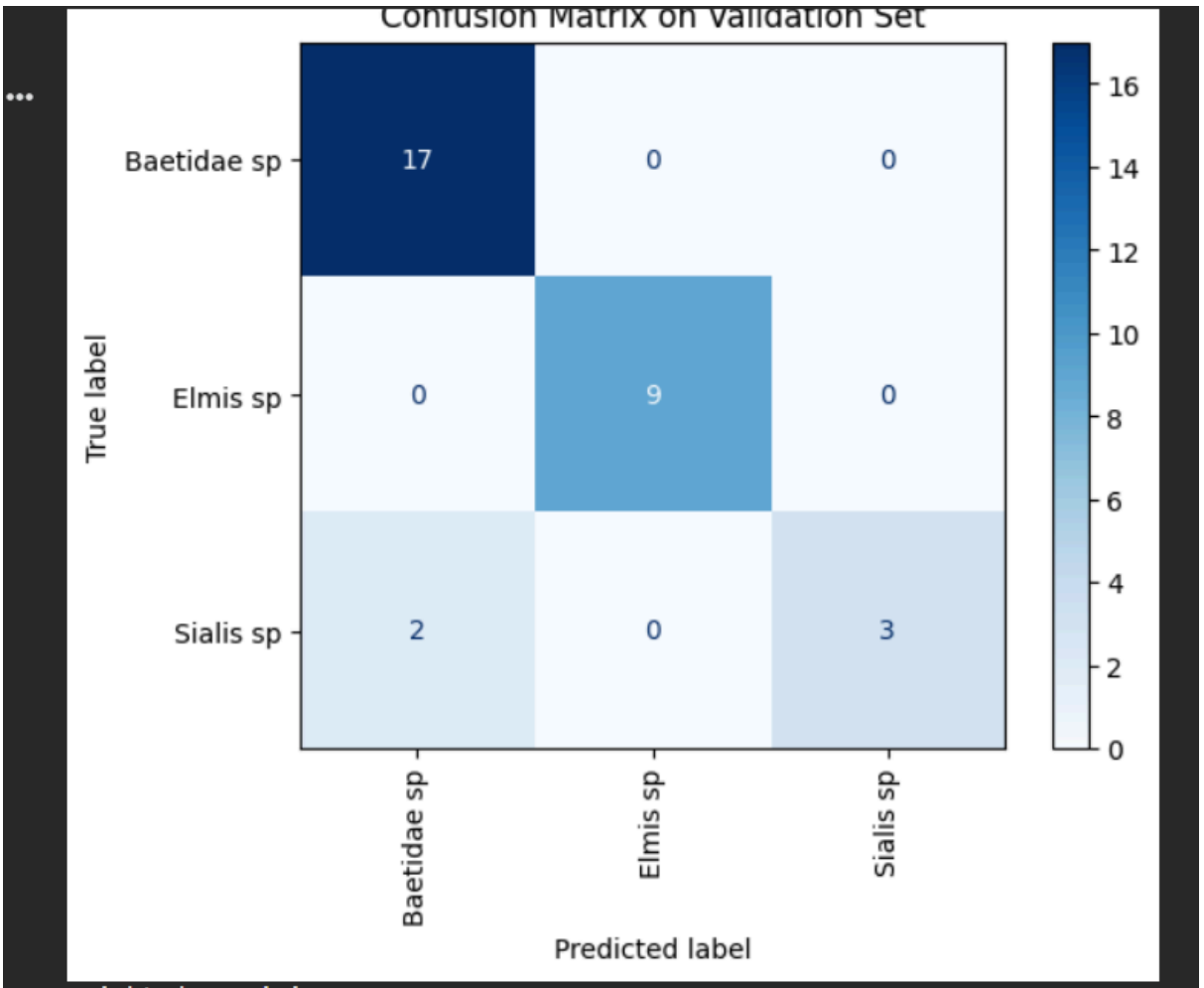
Prints Validation accuracy

Task1_9:

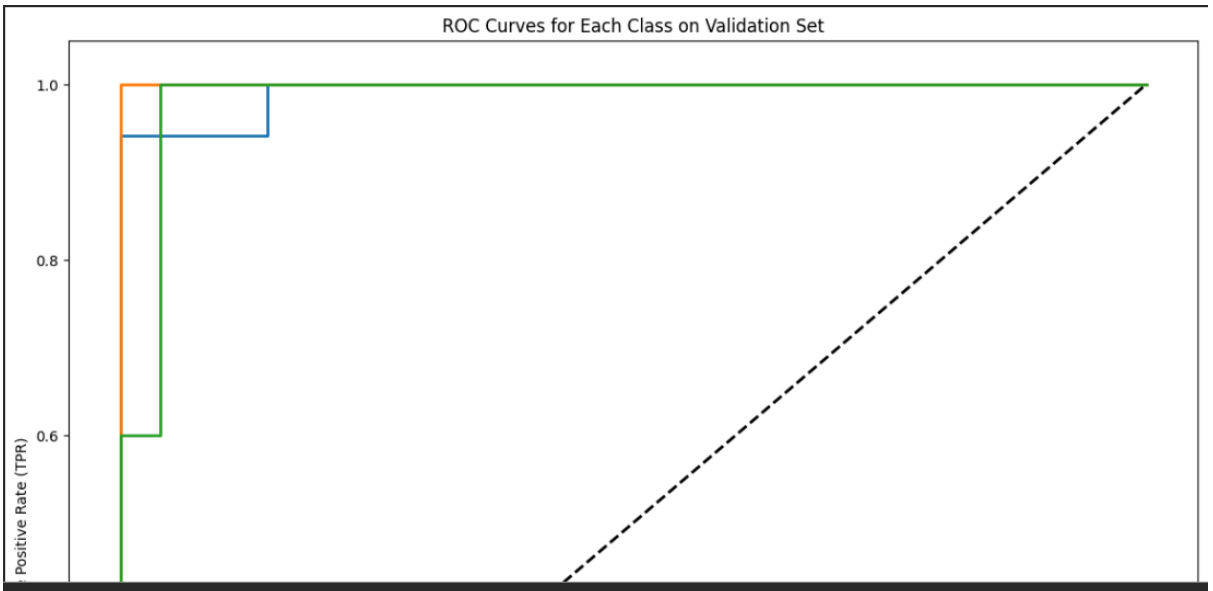
```
/content/drive/MyDrive/Assets/insects_dataset/train_data
Model saved as 'macroinvertebrates_classifier.h5'
```

Saves model as a h5 file.

Task1_10:




```
Weighted Precision: 0.9423
Weighted Recall:    0.9355
Weighted F1-score:  0.9292
```



Displays and graphs confusion matrix and ROC curves